

Programming lang

9/15/2025

HW #1

task 1)

Read a;

Read b;

C := a;

// make copies of a and b

D := b;

L:

// conditional statement to decide where to

if $D > 0$ goto Pos; send value

if $D = 0$ goto end;

goto neg;

// 3 possible cases we can have
(Pos, neg, already 0)

Pos: C := C + 1;

D := D - 1;

goto L;

neg: C := C - 1;

D := D + 1;

goto L;

end:

write X;

halt;

task 2 (BNF / EBNF notation)

$\langle \text{Program} \rangle \rightarrow \langle \text{stmt-list} \rangle \{ \text{stmt-list} \} \text{halt};$

$\langle \text{stmt-list} \rangle \rightarrow [\langle \text{label} \rangle ;] \langle \text{stmt} \rangle ;$

$\langle \text{stmt} \rangle \rightarrow \langle \text{io-stmt} \rangle$

$\quad | \langle \text{assign-stmt} \rangle$

$\quad | \langle \text{if-stmt} \rangle$

$\quad | \langle \text{goto-stmt} \rangle$

$\langle \text{io-stmt} \rangle \rightarrow \text{"read"} \langle \text{var} \rangle$

$\quad \text{"write"} \langle \text{var} \rangle$

$\langle \text{assign-stmt} \rangle \rightarrow \langle \text{var} \rangle := \langle \text{arith-expr} \rangle$

$\langle \text{arith-expr} \rangle \rightarrow \langle \text{var} \rangle$

$\quad | \langle \text{integer} \rangle$

$\quad | \langle \text{var} \rangle + 1$

$\quad | \langle \text{var} \rangle - 1$

$\langle \text{if-stmt} \rangle \rightarrow \text{if } \langle \text{boolean-expr} \rangle \text{ goto } \langle \text{label} \rangle$

$\langle \text{boolean-expr} \rangle \rightarrow \langle \text{var} \rangle \langle \text{relop} \rangle 0$

$\langle \text{relop} \rangle \rightarrow < 1 = 1 >$

$\langle \text{goto-stmt} \rangle \rightarrow \text{goto } \langle \text{label} \rangle$

$\langle \text{var} \rangle \rightarrow \langle \text{lowercase-let} \rangle$

$\langle \text{label} \rangle \rightarrow \langle \text{uppercase-let} \rangle$

$\langle \text{lowercase-let} \rangle \rightarrow \text{"a"} | \text{"b"} | \text{"c"} | \text{"d"} | \text{"e"} | \text{"f"} | \text{"g"} | \text{"h"} | \text{"i"} | \text{"j"} | \text{"k"} | \text{"l"} | \text{"m"} | \text{"n"} | \text{"o"} | \text{"p"} | \text{"q"} | \text{"r"} | \text{"s"} | \text{"t"} | \text{"u"} | \text{"v"} | \text{"w"} | \text{"x"} | \text{"y"} | \text{"z"}$

$\langle \text{uppercase-let} \rangle \rightarrow \text{"A"} | \text{"B"} | \text{"C"} | \text{"D"} | \text{"E"} | \text{"F"} | \text{"G"} | \text{"H"} | \text{"I"} | \text{"J"} | \text{"K"} | \text{"L"} | \text{"M"} | \text{"N"} | \text{"O"} | \text{"P"} | \text{"Q"} | \text{"R"} | \text{"S"} | \text{"T"} | \text{"U"} | \text{"V"} | \text{"W"} | \text{"X"} | \text{"Y"} | \text{"Z"}$

$\langle \text{integer} \rangle \rightarrow [\langle \text{sign} \rangle] \langle \text{digit_sequence} \rangle$

$\langle \text{sign} \rangle \rightarrow "+" | "-"$

$\langle \text{digit_sequence} \rangle \rightarrow 0 \mid \langle \text{nonzero_dig} \rangle \{ \langle \text{digit} \rangle \}$

$\langle \text{digit} \rangle \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

$\langle \text{nonzero_dig} \rangle \rightarrow 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$